

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2016-17

Paper No. CHB-361: Organic & Physical Chemistry

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Organic Chemistry-II – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Define Hückel's rule and show that cycloheptatrienyl cation (tropylium ion) is aromatic.
- (b) Describe Gattermann aldehyde synthesis of benzaldehyde.
- (c) Predict the products when benzaldehyde is reacted with: (i) Conc. KOH and (ii) Acetophenone in dilute NaOH.
- (d) Give the synthesis and mechanism of phenolphthalein from phthalic anhydride and phenol in the presence of H₂SO₄.
- (e) Explain why pyridine is a stronger base than pyrrole.

Question 2.

- (a) Outline the synthesis of Alizarin and Indigo dyes starting from phthalic anhydride. Give structural representations. [6]
- (b) Write short notes on: [6]
 - (i) Bischler-Napieralski synthesis of isoquinoline
 - (ii) Reimer-Tiemann reaction mechanism

Question 3.

- (a) Discuss electrophilic aromatic substitution in naphthalene. Why does substitution occur preferentially at the α -position? [6]
- (b) How will you synthesize: (i) Malachite green and (ii) Fluorescein dyes? [6]

SECTION B

(Physical Chemistry-I – Marks: 35)

Question 4. Answer all parts of the following: [5 × 2 = 10]

- (a) Sketch the diagram of a conductivity cell and define cell constant.
 - (b) State Kohlrausch's law of independent migration of ions.
 - (c) Explain why the equivalent conductance of a strong electrolyte increases with dilution.
 - (d) Give a brief definition of transport number of an ion.
 - (e) Define the term ionic mobility and state its units.
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Question 5.

- (a) Describe the moving boundary method for the determination of transport numbers of ions. [6]
 - (b) The resistance of a 0.1 M KCl solution in a cell was found to be 85 ohm (specific conductivity of 0.1 M KCl = $0.01289 \text{ ohm}^{-1} \text{ cm}^{-1}$). Calculate the cell constant. If the resistance of a 0.05 M solution of another electrolyte in the same cell is 220 ohm, calculate its specific and molar conductivity. [6]
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Question 6.

- (a) Discuss the Debye-Hückel-Onsager equation for strong electrolytes and explain the relaxation and electrophoretic effects. [8]
 - (b) Explain conductometric titrations of: (i) HCl vs. NaOH and (ii) CH₃COOH vs. NaOH. [4]
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